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Introduction to Document Processing

This is a sample PDF document created for testing the document processor.

Document processing is a critical component of RAG systems.

Efficient extraction of text from PDFs ensures data quality.

The document processor supports both PDF and TXT file formats.

Character-based chunking allows flexible control over chunk size and overlap.

Metadata is automatically extracted for each chunk.

Key Features of M1:

- PDF text extraction with pdfplumber
- TXT file ingestion with UTF-8 encoding support
- Character-based chunking with configurable size and overlap
- Automatic metadata extraction
- Stable chunk ID generation based on filename and sequence
- JSON output for easy integration with downstream components

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Advanced Features

Error handling ensures robustness when processing files.
The processor gracefully skips problematic files and continues.
Chunk overlap preservation ensures context is maintained.
The char_start and char_end positions enable reconstruction.

Testing and Validation:

- Comprehensive unit tests validate all M1 functionality
- Sample documents are generated automatically for testing
- Edge cases are handled including empty folders
- Statistical reporting provides insights into chunking results

Performance Characteristics:

- Character-based chunking is fast and deterministic
- Memory usage scales linearly with document size
- No external dependencies for core chunking logic
- Ready for integration with embeddings in M2